

# AviLight Ecological Impact and Diagnostics Report

## Executive Summary

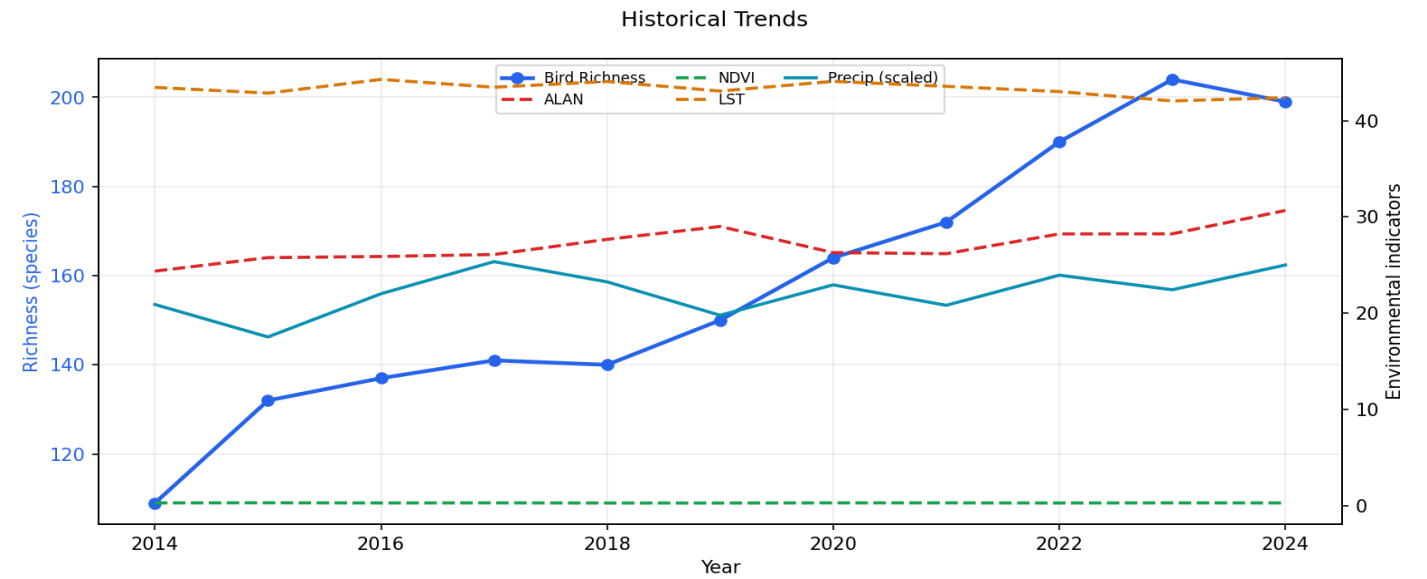
### Run Context

| Field       | Value                   |
|-------------|-------------------------|
| Area        | All Areas               |
| Trend Years | 2014-2024               |
| Snapshot    | 2024-01                 |
| Generated   | 2026-04-18 20:43:50 UTC |

# AviLight Ecological Impact and Diagnostics Report

## Ecological Impact View

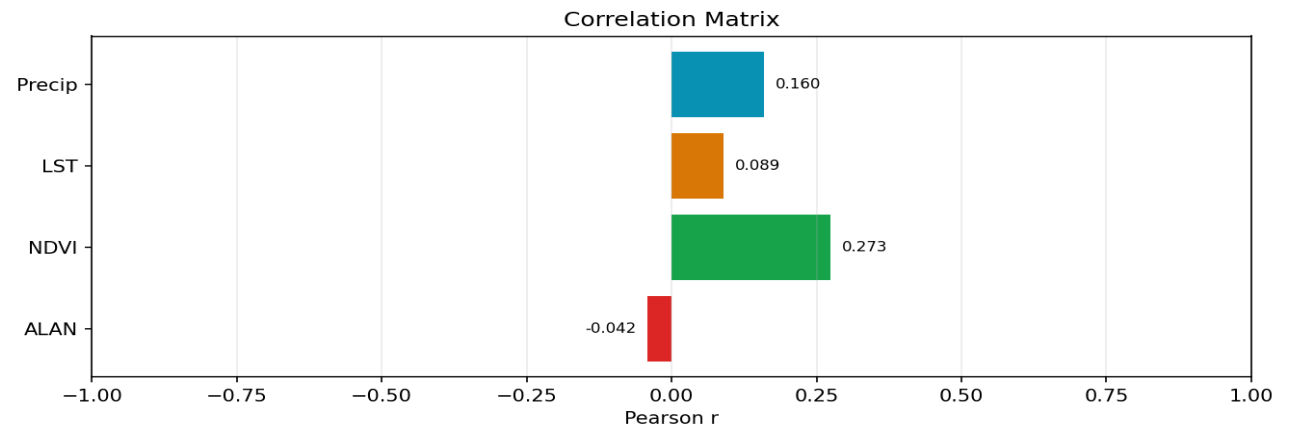
Chart: Long-Term Ecological Impact Trends



Numerical Values: Historical Trends

| Year | Richness | ALAN  | NDVI   | LST   | Precip  |
|------|----------|-------|--------|-------|---------|
| 2014 | 109.00   | 24.36 | 0.2843 | 43.45 | 2089.67 |
| 2015 | 132.00   | 25.77 | 0.2935 | 42.86 | 1752.42 |
| 2016 | 137.00   | 25.88 | 0.2766 | 44.28 | 2201.25 |
| 2017 | 141.00   | 26.09 | 0.2854 | 43.47 | 2535.23 |
| 2018 | 140.00   | 27.66 | 0.2721 | 44.06 | 2324.73 |
| 2019 | 150.00   | 29.00 | 0.2682 | 43.07 | 1977.59 |
| 2020 | 164.00   | 26.29 | 0.2856 | 44.08 | 2293.80 |
| 2021 | 172.00   | 26.18 | 0.2857 | 43.56 | 2081.15 |
| 2022 | 190.00   | 28.23 | 0.2724 | 43.02 | 2394.15 |
| 2023 | 204.00   | 28.24 | 0.2864 | 42.05 | 2242.81 |
| 2024 | 199.00   | 30.66 | 0.2810 | 42.39 | 2500.04 |

Chart: Correlation Matrix

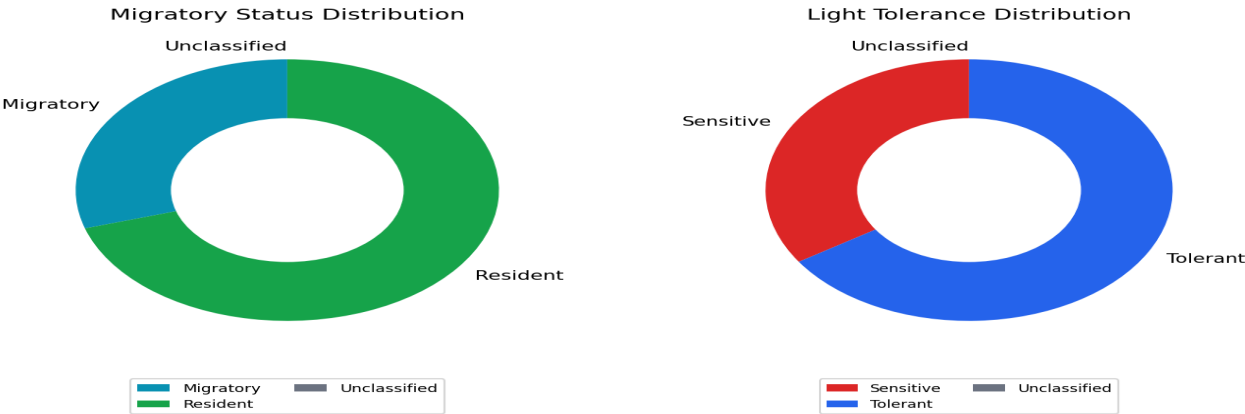


# AviLight Ecological Impact and Diagnostics Report

## Numerical Values: Correlation Coefficients

| Metric                    | Pearson r |
|---------------------------|-----------|
| ALAN vs Richness          | -0.0415   |
| NDVI vs Richness          | 0.2732    |
| LST vs Richness           | 0.0890    |
| Precipitation vs Richness | 0.1595    |

Chart: Migratory and Light Tolerance Distribution



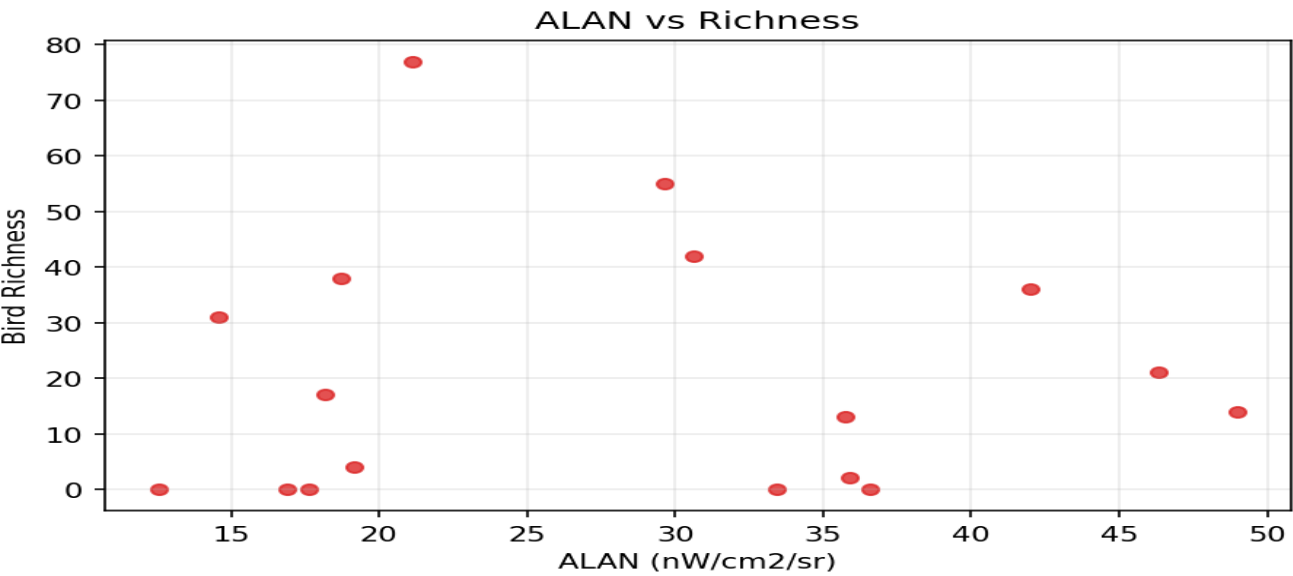
## Numerical Values: Migration Distribution

| Category     | Count |
|--------------|-------|
| Migratory    | 33    |
| Resident     | 78    |
| Unclassified | 0     |

## Numerical Values: Light Tolerance Distribution

| Category     | Count |
|--------------|-------|
| Sensitive    | 38    |
| Tolerant     | 73    |
| Unclassified | 0     |

Chart: ALAN vs Richness

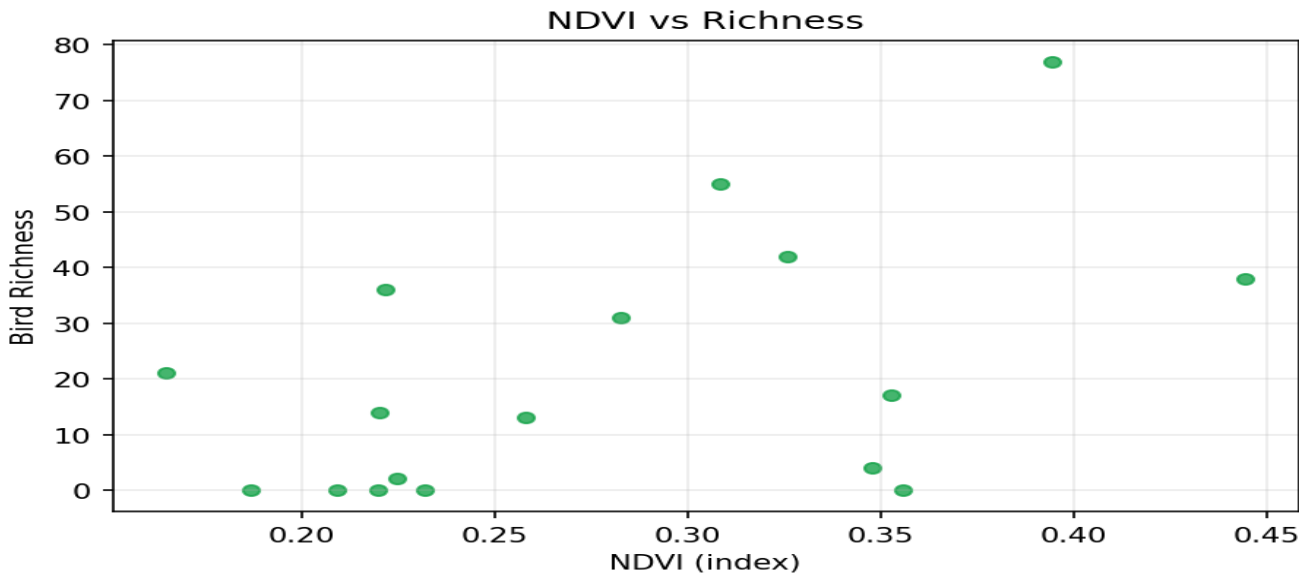


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Numerical Values: ALAN vs Richness

| Site        | Area        | X       | Richness |
|-------------|-------------|---------|----------|
| Caloocan    | Caloocan    | 17.6253 | 0.00     |
| Las Piñas   | Las Piñas   | 18.1804 | 17.00    |
| Makati      | Makati      | 42.0139 | 36.00    |
| Malabon     | Malabon     | 16.8709 | 0.00     |
| Mandaluyong | Mandaluyong | 35.8872 | 2.00     |
| Manila      | Manila      | 46.3457 | 21.00    |
| Marikina    | Marikina    | 19.1350 | 4.00     |
| Muntinlupa  | Muntinlupa  | 18.7014 | 38.00    |
| Navotas     | Navotas     | 12.5424 | 0.00     |
| Parañaque   | Parañaque   | 29.6273 | 55.00    |
| Pasay       | Pasay       | 48.9901 | 14.00    |
| Pasig       | Pasig       | 35.7328 | 13.00    |
| Pateros     | Pateros     | 33.4426 | 0.00     |
| Quezon City | Quezon City | 21.1494 | 77.00    |
| San Juan    | San Juan    | 36.5872 | 0.00     |
| Taguig      | Taguig      | 30.6320 | 42.00    |
| Valenzuela  | Valenzuela  | 14.5767 | 31.00    |

Chart: NDVI vs Richness



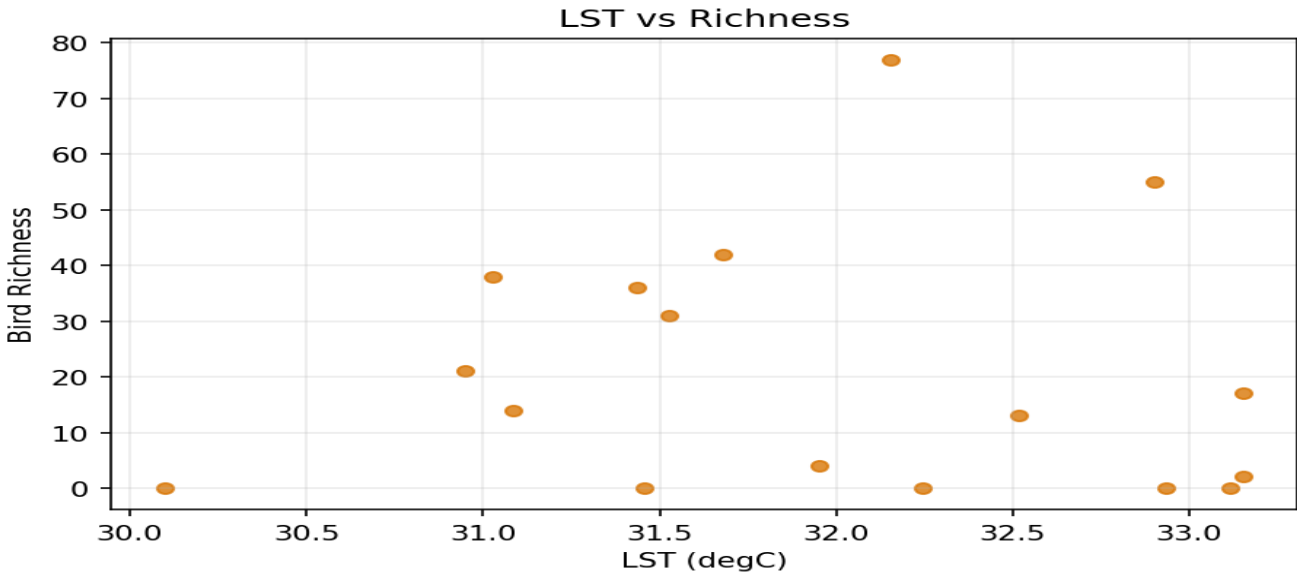
Numerical Values: NDVI vs Richness

| Site        | Area        | X      | Richness |
|-------------|-------------|--------|----------|
| Caloocan    | Caloocan    | 0.3558 | 0.00     |
| Las Piñas   | Las Piñas   | 0.3528 | 17.00    |
| Makati      | Makati      | 0.2215 | 36.00    |
| Malabon     | Malabon     | 0.2092 | 0.00     |
| Mandaluyong | Mandaluyong | 0.2247 | 2.00     |
| Manila      | Manila      | 0.1650 | 21.00    |
| Marikina    | Marikina    | 0.3479 | 4.00     |
| Muntinlupa  | Muntinlupa  | 0.4443 | 38.00    |
| Navotas     | Navotas     | 0.1869 | 0.00     |

AviLight Ecological Impact and Diagnostics Report

|             |             |        |       |
|-------------|-------------|--------|-------|
| Parañaque   | Parañaque   | 0.3084 | 55.00 |
| Pasay       | Pasay       | 0.2203 | 14.00 |
| Pasig       | Pasig       | 0.2579 | 13.00 |
| Pateros     | Pateros     | 0.2320 | 0.00  |
| Quezon City | Quezon City | 0.3943 | 77.00 |
| San Juan    | San Juan    | 0.2197 | 0.00  |
| Taguig      | Taguig      | 0.3258 | 42.00 |
| Valenzuela  | Valenzuela  | 0.2828 | 31.00 |

Chart: LST vs Richness

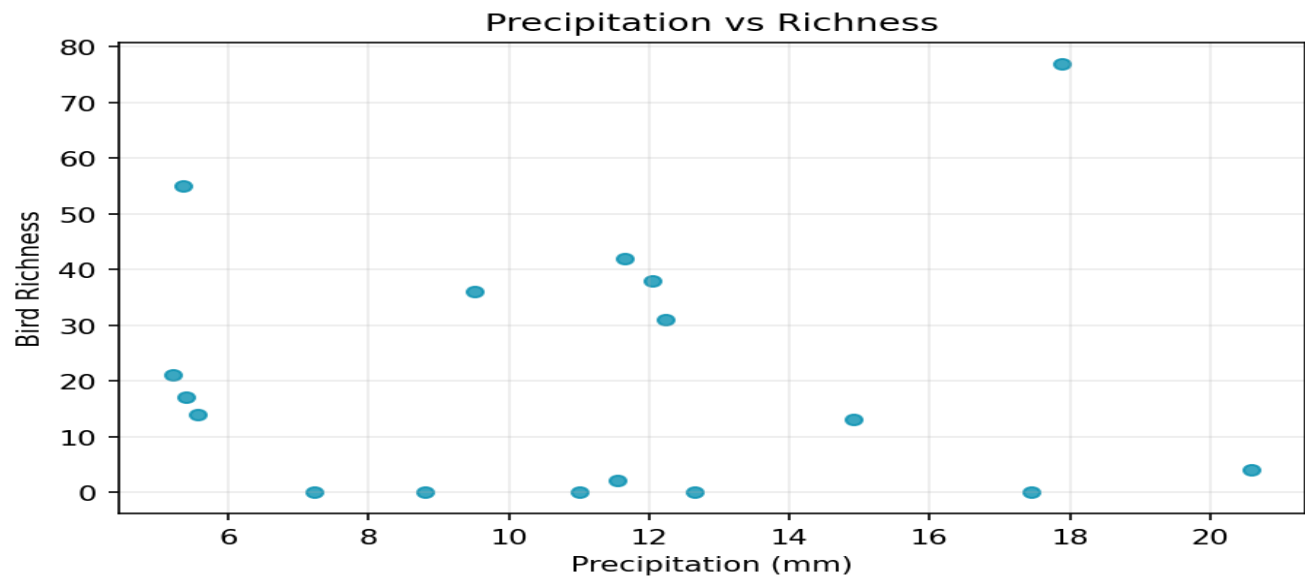


Numerical Values: LST vs Richness

| Site        | Area        | X       | Richness |
|-------------|-------------|---------|----------|
| Caloocan    | Caloocan    | 31.4546 | 0.00     |
| Las Piñas   | Las Piñas   | 33.1531 | 17.00    |
| Makati      | Makati      | 31.4346 | 36.00    |
| Malabon     | Malabon     | 30.0974 | 0.00     |
| Mandaluyong | Mandaluyong | 33.1521 | 2.00     |
| Manila      | Manila      | 30.9494 | 21.00    |
| Marikina    | Marikina    | 31.9531 | 4.00     |
| Muntinlupa  | Muntinlupa  | 31.0285 | 38.00    |
| Navotas     | Navotas     | 32.9352 | 0.00     |
| Parañaque   | Parañaque   | 32.9000 | 55.00    |
| Pasay       | Pasay       | 31.0866 | 14.00    |
| Pasig       | Pasig       | 32.5155 | 13.00    |
| Pateros     | Pateros     | 32.2433 | 0.00     |
| Quezon City | Quezon City | 32.1529 | 77.00    |
| San Juan    | San Juan    | 33.1166 | 0.00     |
| Taguig      | Taguig      | 31.6808 | 42.00    |
| Valenzuela  | Valenzuela  | 31.5286 | 31.00    |

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Chart: Precipitation vs Richness

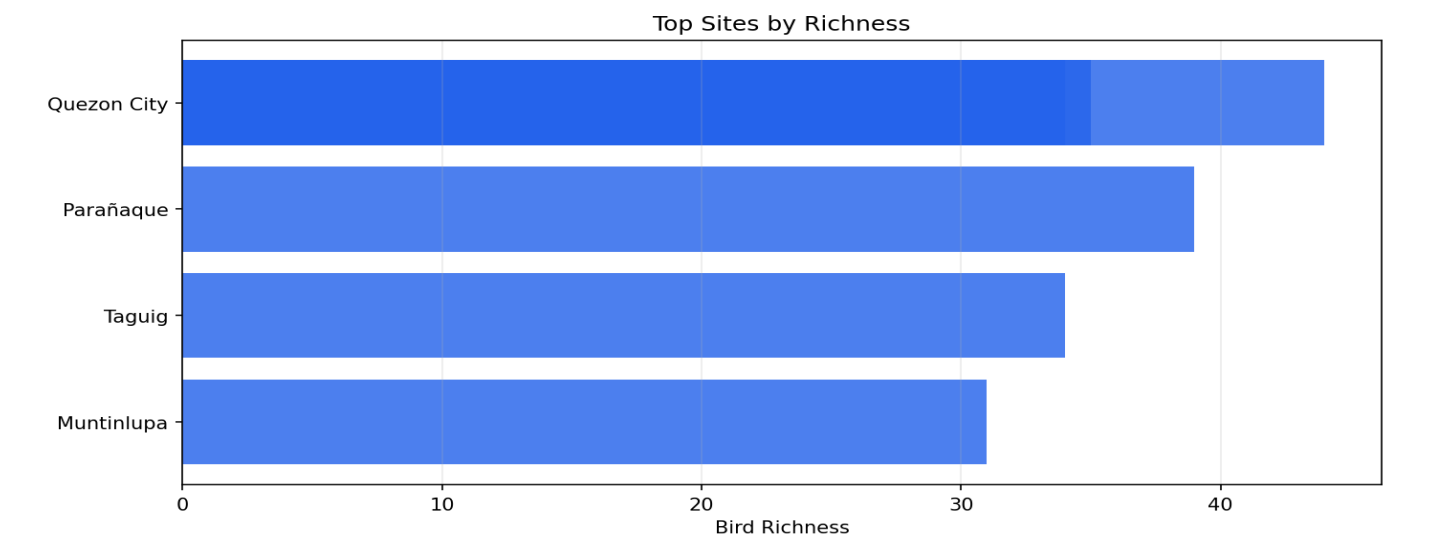


Numerical Values: Precipitation vs Richness

| Site        | Area        | X       | Richness |
|-------------|-------------|---------|----------|
| Caloocan    | Caloocan    | 17.4396 | 0.00     |
| Las Piñas   | Las Piñas   | 5.3914  | 17.00    |
| Makati      | Makati      | 9.5073  | 36.00    |
| Malabon     | Malabon     | 8.8099  | 0.00     |
| Mandaluyong | Mandaluyong | 11.5377 | 2.00     |
| Manila      | Manila      | 5.1975  | 21.00    |
| Marikina    | Marikina    | 20.5884 | 4.00     |
| Muntinlupa  | Muntinlupa  | 12.0530 | 38.00    |
| Navotas     | Navotas     | 7.2133  | 0.00     |
| Parañaque   | Parañaque   | 5.3558  | 55.00    |
| Pasay       | Pasay       | 5.5558  | 14.00    |
| Pasig       | Pasig       | 14.9217 | 13.00    |
| Pateros     | Pateros     | 12.6396 | 0.00     |
| Quezon City | Quezon City | 17.8904 | 77.00    |
| San Juan    | San Juan    | 11.0117 | 0.00     |
| Taguig      | Taguig      | 11.6590 | 42.00    |
| Valenzuela  | Valenzuela  | 12.2234 | 31.00    |

# AviLight Ecological Impact and Diagnostics Report

Chart: Top 10 Sites by Richness



Numerical Values: Top Sites by Richness

| Rank | Area        | Richness |
|------|-------------|----------|
| 1    | Quezon City | 44       |
| 2    | Parañaque   | 39       |
| 3    | Quezon City | 35       |
| 4    | Quezon City | 34       |
| 5    | Taguig      | 34       |
| 6    | Quezon City | 34       |
| 7    | Quezon City | 32       |
| 8    | Quezon City | 32       |
| 9    | Quezon City | 31       |
| 10   | Muntinlupa  | 31       |

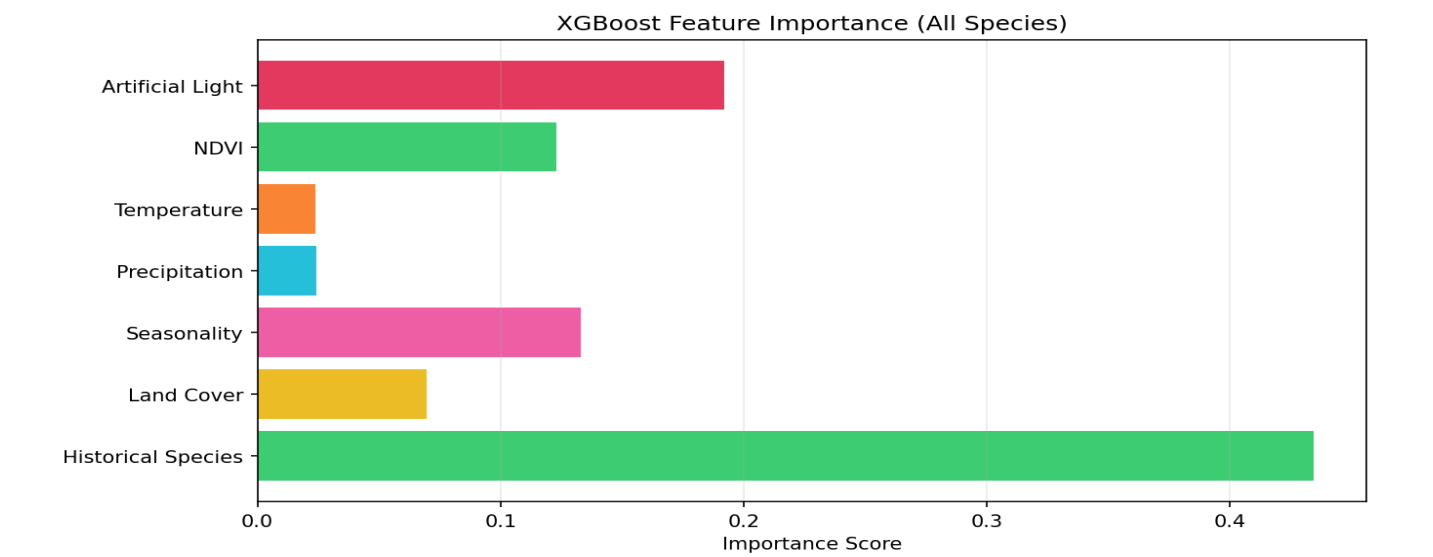
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## Model Diagnostics View

### KPI Cards

| Metric | Value  |
|--------|--------|
| RMSE   | 6.7345 |
| MAE    | 4.1445 |
| R2     | 0.7061 |

Chart: XGBoost Feature Importance (All Species)

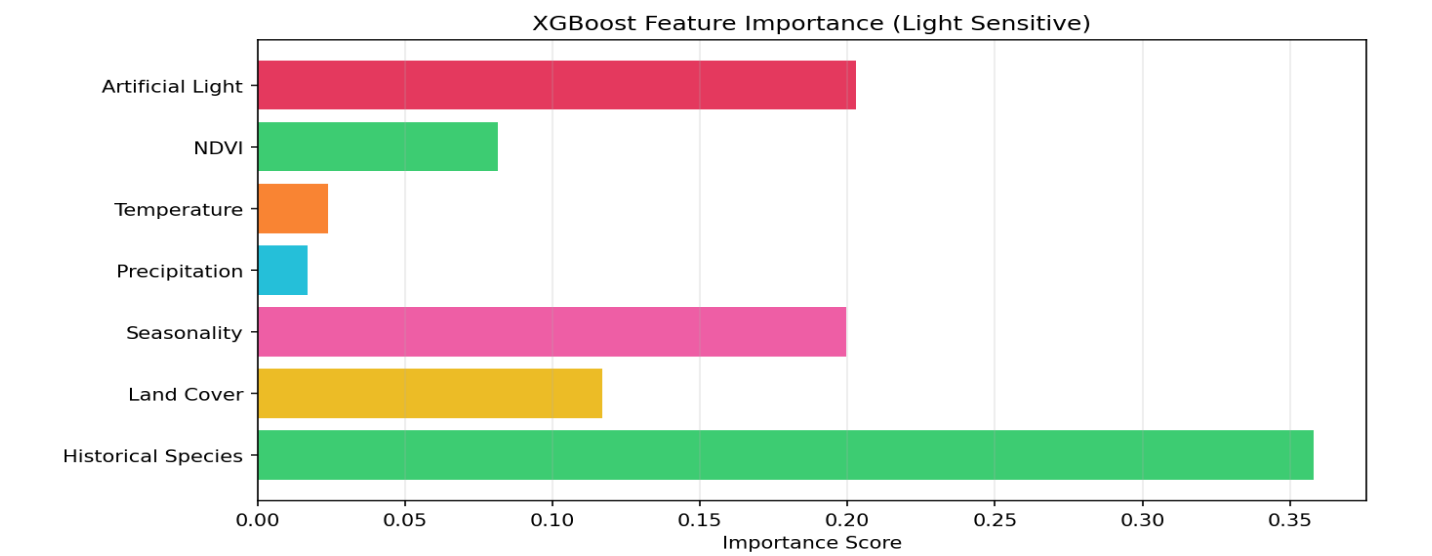


### Numerical Values: XGBoost Feature Importance (All Species)

| Feature            | Importance |
|--------------------|------------|
| Artificial Light   | 0.191991   |
| NDVI               | 0.122968   |
| Temperature        | 0.023987   |
| Precipitation      | 0.024173   |
| Seasonality        | 0.133136   |
| Land Cover         | 0.069444   |
| Historical Species | 0.434300   |

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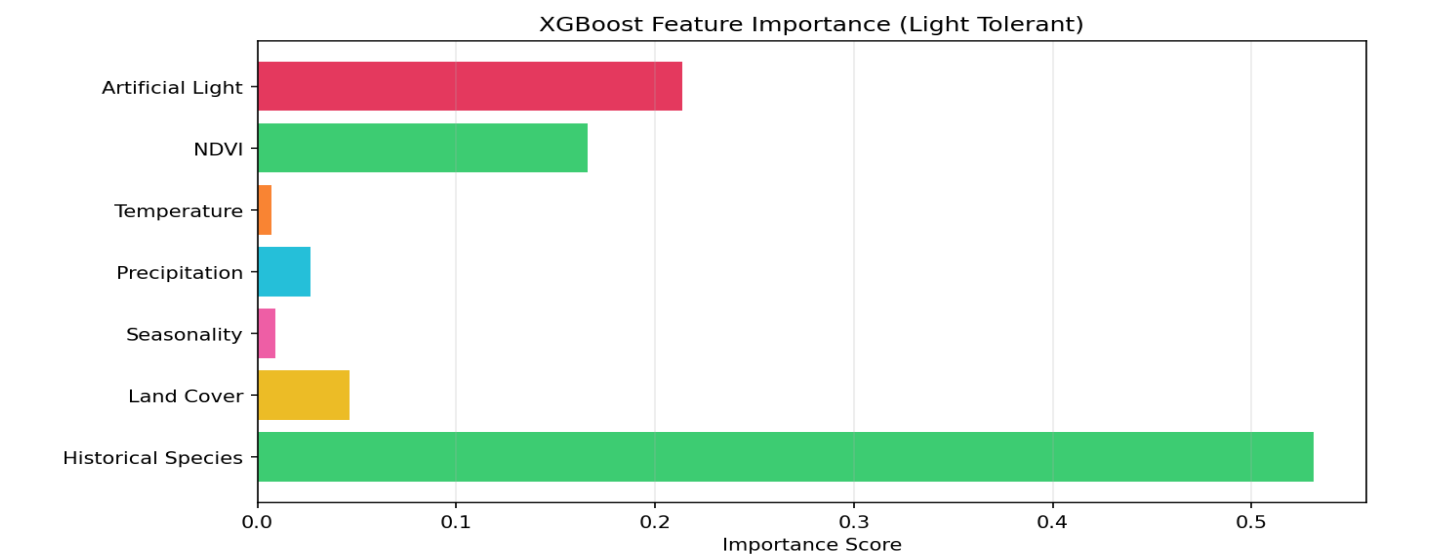
Chart: XGBoost Feature Importance (Light Sensitive)



Numerical Values: XGBoost Feature Importance (Light Sensitive)

| Feature            | Importance |
|--------------------|------------|
| Artificial Light   | 0.202727   |
| NDVI               | 0.081639   |
| Temperature        | 0.024060   |
| Precipitation      | 0.017154   |
| Seasonality        | 0.199484   |
| Land Cover         | 0.117012   |
| Historical Species | 0.357924   |

Chart: XGBoost Feature Importance (Light Tolerant)



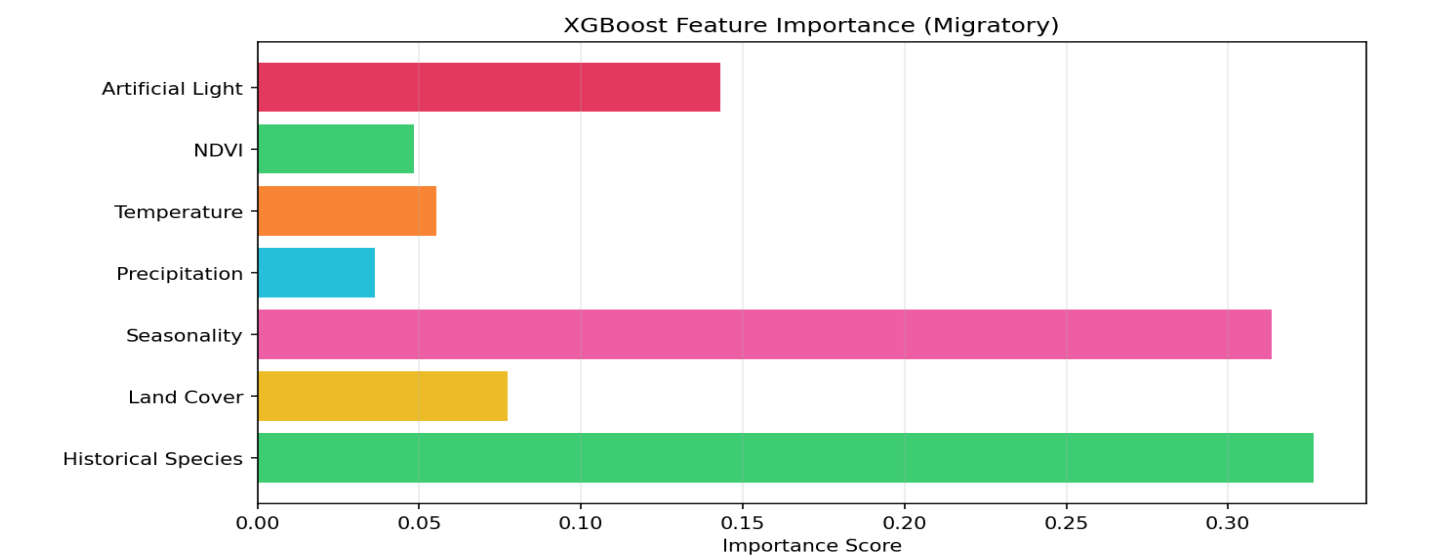
Numerical Values: XGBoost Feature Importance (Light Tolerant)

| Feature          | Importance |
|------------------|------------|
| Artificial Light | 0.213590   |
| NDVI             | 0.166192   |
| Temperature      | 0.007116   |

# AviLight Ecological Impact and Diagnostics Report

|                    |          |
|--------------------|----------|
| Precipitation      | 0.026545 |
| Seasonality        | 0.009277 |
| Land Cover         | 0.046179 |
| Historical Species | 0.531101 |

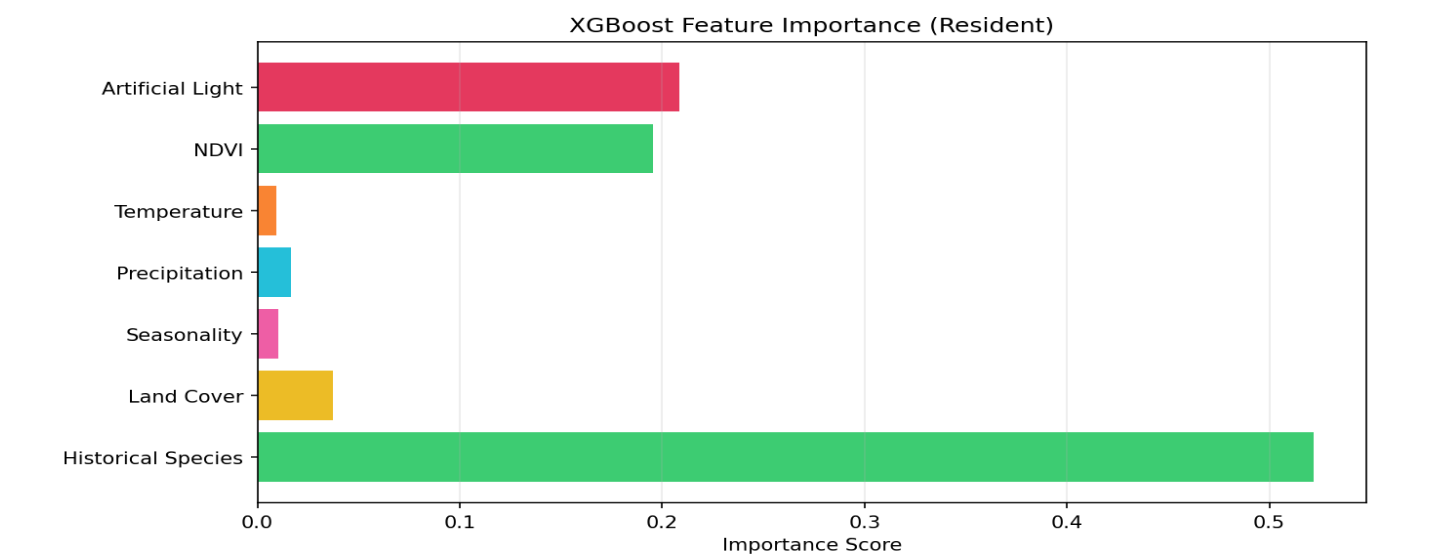
Chart: XGBoost Feature Importance (Migratory)



Numerical Values: XGBoost Feature Importance (Migratory)

| Feature            | Importance |
|--------------------|------------|
| Artificial Light   | 0.143060   |
| NDVI               | 0.048403   |
| Temperature        | 0.055268   |
| Precipitation      | 0.036267   |
| Seasonality        | 0.313430   |
| Land Cover         | 0.077193   |
| Historical Species | 0.326379   |

Chart: XGBoost Feature Importance (Resident)

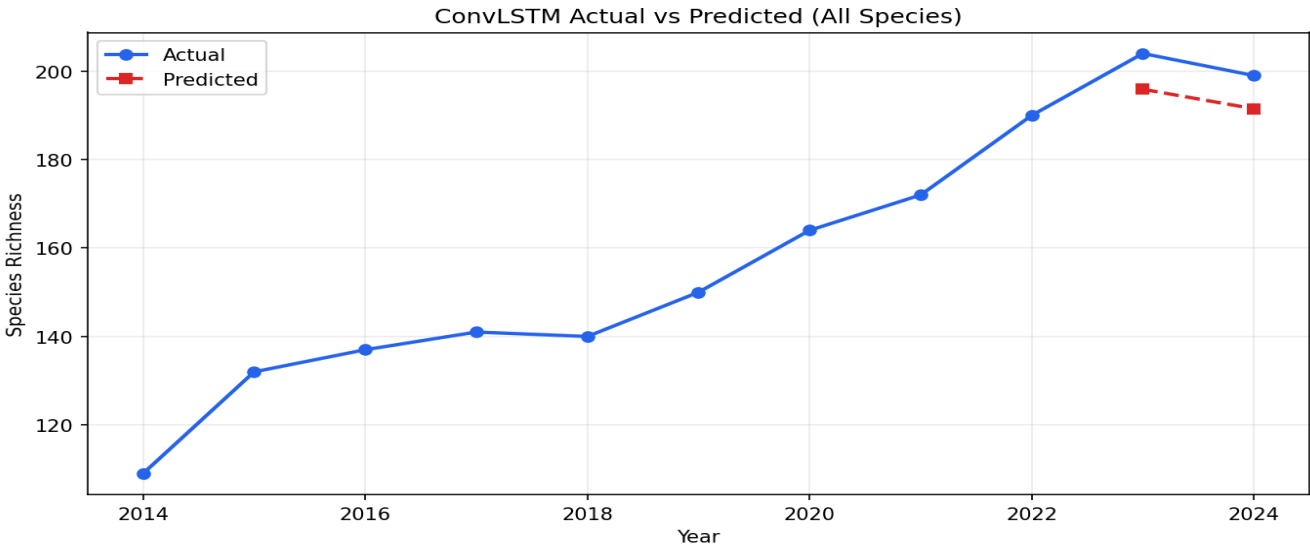


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## Numerical Values: XGBoost Feature Importance (Resident)

| Feature            | Importance |
|--------------------|------------|
| Artificial Light   | 0.208588   |
| NDVI               | 0.195638   |
| Temperature        | 0.009504   |
| Precipitation      | 0.016725   |
| Seasonality        | 0.010355   |
| Land Cover         | 0.037393   |
| Historical Species | 0.521797   |

Chart: ConvLSTM Actual vs Predicted (All Species)

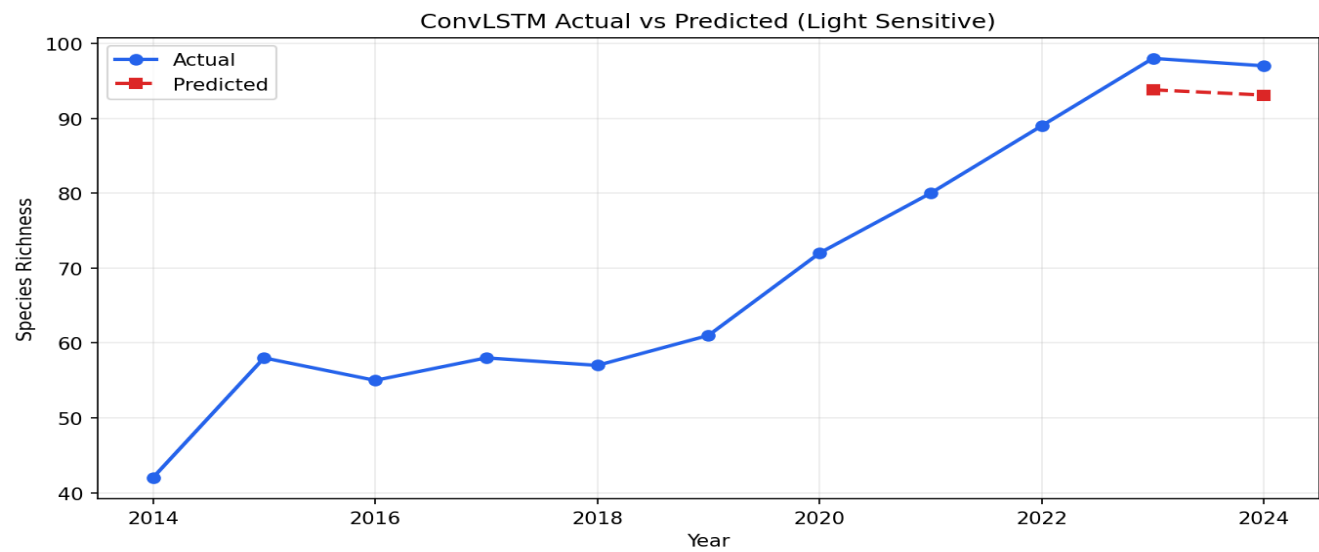


## Numerical Values: ConvLSTM Actual vs Predicted (All Species)

| Year | Actual   | Predicted |
|------|----------|-----------|
| 2014 | 109.0000 | -         |
| 2015 | 132.0000 | -         |
| 2016 | 137.0000 | -         |
| 2017 | 141.0000 | -         |
| 2018 | 140.0000 | -         |
| 2019 | 150.0000 | -         |
| 2020 | 164.0000 | -         |
| 2021 | 172.0000 | -         |
| 2022 | 190.0000 | -         |
| 2023 | 204.0000 | 195.9000  |
| 2024 | 199.0000 | 191.5000  |

# AviLight Ecological Impact and Diagnostics Report

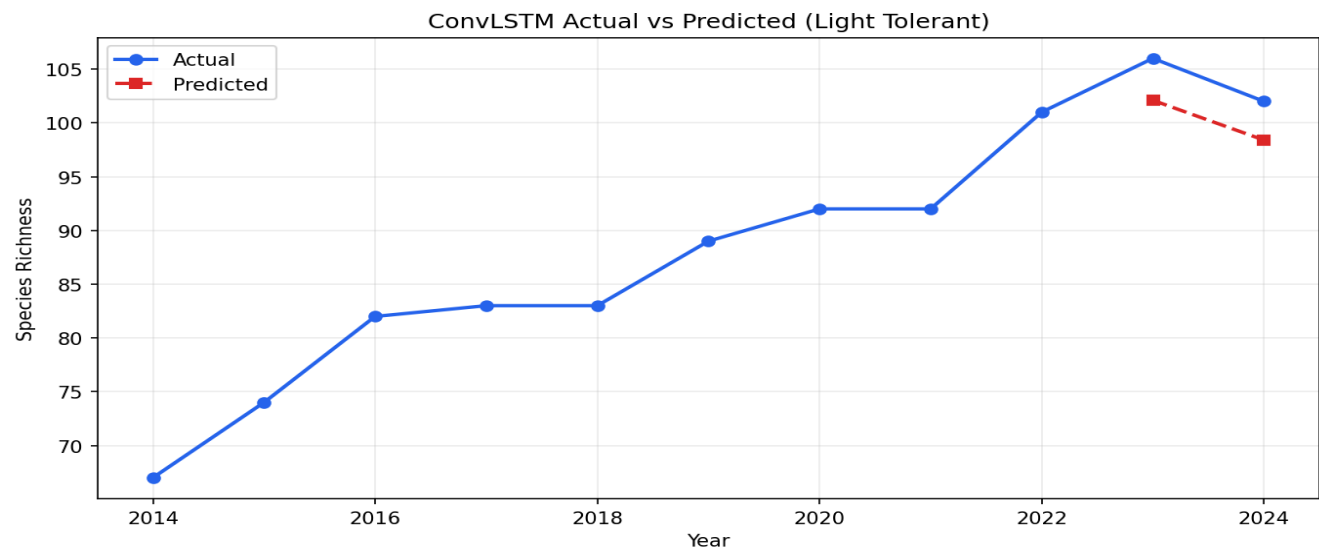
Chart: ConvLSTM Actual vs Predicted (Light Sensitive)



Numerical Values: ConvLSTM Actual vs Predicted (Light Sensitive)

| Year |      | Actual  | Predicted |
|------|------|---------|-----------|
|      | 2014 | 42.0000 | -         |
|      | 2015 | 58.0000 | -         |
|      | 2016 | 55.0000 | -         |
|      | 2017 | 58.0000 | -         |
|      | 2018 | 57.0000 | -         |
|      | 2019 | 61.0000 | -         |
|      | 2020 | 72.0000 | -         |
|      | 2021 | 80.0000 | -         |
|      | 2022 | 89.0000 | -         |
|      | 2023 | 98.0000 | 93.8000   |
|      | 2024 | 97.0000 | 93.1000   |

Chart: ConvLSTM Actual vs Predicted (Light Tolerant)

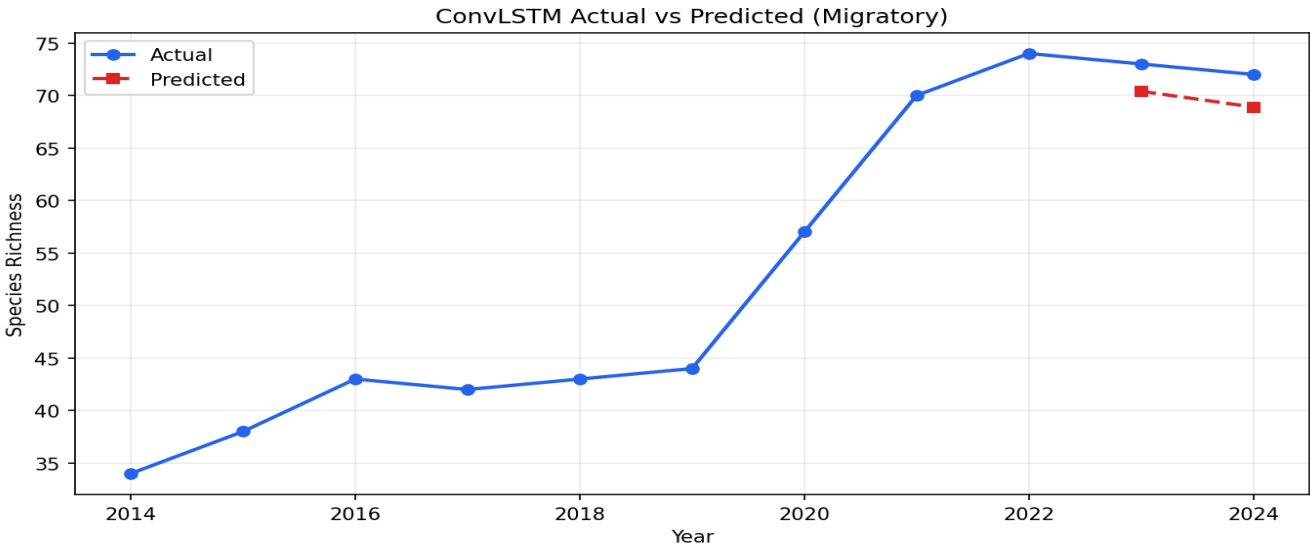


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Numerical Values: ConvLSTM Actual vs Predicted (Light Tolerant)

| Year | Actual   | Predicted |
|------|----------|-----------|
| 2014 | 67.0000  | -         |
| 2015 | 74.0000  | -         |
| 2016 | 82.0000  | -         |
| 2017 | 83.0000  | -         |
| 2018 | 83.0000  | -         |
| 2019 | 89.0000  | -         |
| 2020 | 92.0000  | -         |
| 2021 | 92.0000  | -         |
| 2022 | 101.0000 | -         |
| 2023 | 106.0000 | 102.1000  |
| 2024 | 102.0000 | 98.4000   |

Chart: ConvLSTM Actual vs Predicted (Migratory)

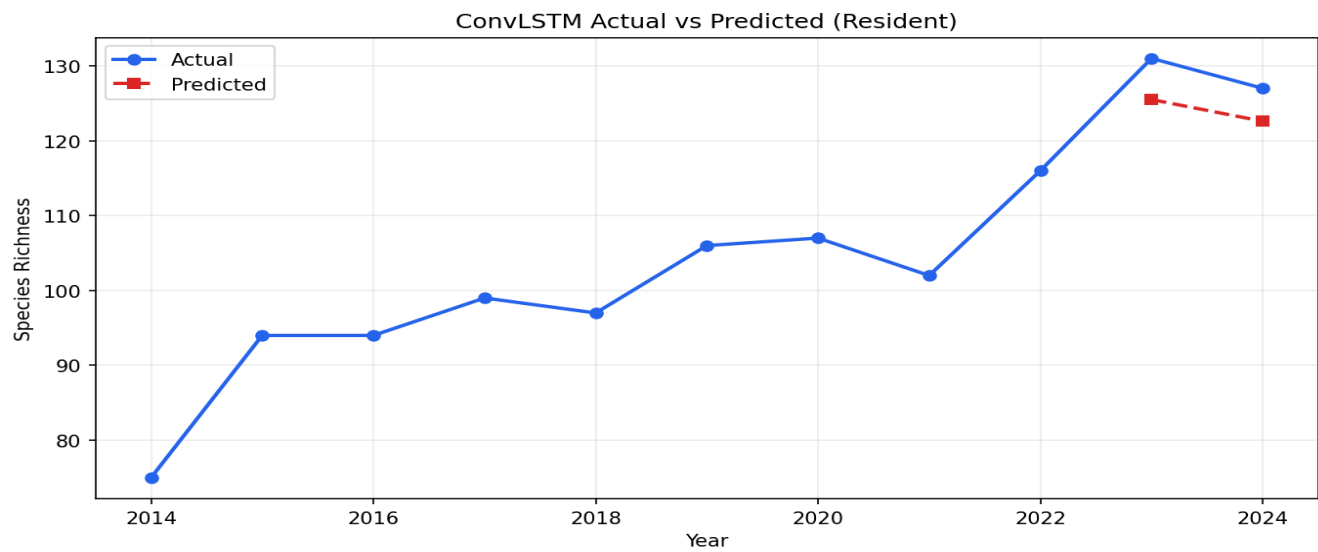


Numerical Values: ConvLSTM Actual vs Predicted (Migratory)

| Year | Actual  | Predicted |
|------|---------|-----------|
| 2014 | 34.0000 | -         |
| 2015 | 38.0000 | -         |
| 2016 | 43.0000 | -         |
| 2017 | 42.0000 | -         |
| 2018 | 43.0000 | -         |
| 2019 | 44.0000 | -         |
| 2020 | 57.0000 | -         |
| 2021 | 70.0000 | -         |
| 2022 | 74.0000 | -         |
| 2023 | 73.0000 | 70.4000   |
| 2024 | 72.0000 | 68.9000   |

# AviLight Ecological Impact and Diagnostics Report

Chart: ConvLSTM Actual vs Predicted (Resident)



Numerical Values: ConvLSTM Actual vs Predicted (Resident)

| Year | Actual   | Predicted |
|------|----------|-----------|
| 2014 | 75.0000  | -         |
| 2015 | 94.0000  | -         |
| 2016 | 94.0000  | -         |
| 2017 | 99.0000  | -         |
| 2018 | 97.0000  | -         |
| 2019 | 106.0000 | -         |
| 2020 | 107.0000 | -         |
| 2021 | 102.0000 | -         |
| 2022 | 116.0000 | -         |
| 2023 | 131.0000 | 125.5000  |
| 2024 | 127.0000 | 122.6000  |

Systems Recommendations: KBA/PA Audit

| Site                                 | Type | Species | Sensitive% | Light | Eff.Score |
|--------------------------------------|------|---------|------------|-------|-----------|
| La Mesa Watershed                    | KBA  | 85      | 42.0       | 28.5  | 82.0      |
| Ninoy Aquino Parks & Wildlife Center | PA   | 48      | 35.0       | 45.2  | 68.0      |
| Las Piñas-Parañaque Critical Habitat | KBA  | 92      | 58.0       | 38.7  | 75.0      |
| Marikina Watershed                   | PA   | 72      | 48.0       | 32.1  | 79.0      |
| Laguna de Bay Wetlands               | KBA  | 125     | 52.0       | 35.4  | 71.0      |